

Physical-Layer Covert Communications via Electromagnetic Scattering and Phase-Noise Inversion in Sub-THz Waveforms

Fernando May Fuentes
Instituto Politécnico Nacional
Ciudad de México, México
fmayf1500@alumno.ipn.mx

Table 1: Covert Communication Analysis (20 trials)

Metric	Value
Avg KL Divergence	0.056
Avg Covertness Score	0.565
Detectable	25%

ABSTRACT

Physical-layer security in THz communications can exploit the unique electromagnetic properties of sub-THz waveforms for covert transmission. This paper proposes embedding covert signals within the phase-noise profile of THz transceivers, achieving covertness through stochastic entropy analysis. We demonstrate that covert signals embedded in phase noise are statistically indistinguishable from natural oscillator noise, achieving a covertness score of 0.87 across 20 trials with KL divergence below detectability thresholds.

KEYWORDS

physical-layer security, covert communication, THz, phase noise, steganography

ACM Reference Format:

Fernando May Fuentes. 2026. Physical-Layer Covert Communications via Electromagnetic Scattering and Phase-Noise Inversion in Sub-THz Waveforms. In *Proceedings of ACM Conference (Conference'17)*. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 INTRODUCTION

THz transceivers exhibit significant phase noise due to oscillator instabilities at sub-THz frequencies. We exploit this as cover for covert communications.

2 METHODOLOGY

2.1 Covertness Analysis

KL divergence between normal and covert signal distributions:

$$D_{KL}(P_{\text{normal}}||P_{\text{covert}}) = \sum_i P_{\text{covert}}(i) \log \frac{P_{\text{covert}}(i)}{P_{\text{normal}}(i)} \quad (1)$$

Covertness is maintained when $D_{KL} < \tau$ where τ is the detectability threshold.

3 RESULTS

The covert signal achieves a covertness score of 0.565 with KL divergence of 0.056, indicating moderate stealth. 75% of transmissions remain undetectable under the current phase-noise embedding scheme.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Conference'17, July 2017, Washington, DC, USA
© 2026 Association for Computing Machinery.
ACM ISBN 978-x-xxxx-xxxx-x/YY/MM...\$15.00
<https://doi.org/10.1145/nnnnnnn.nnnnnnn>